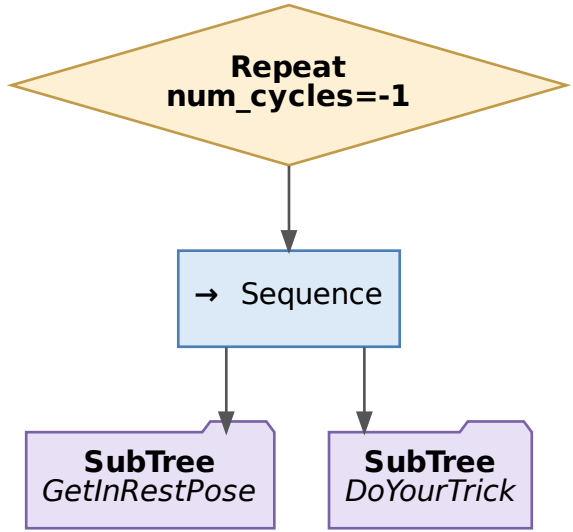
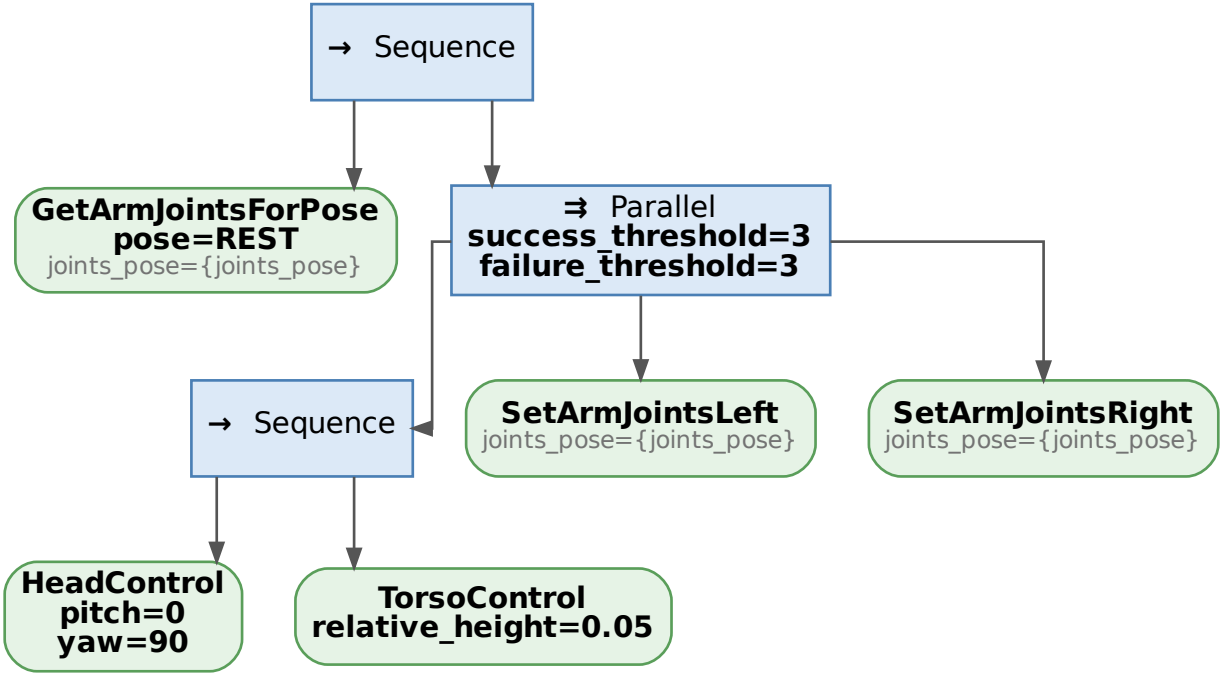
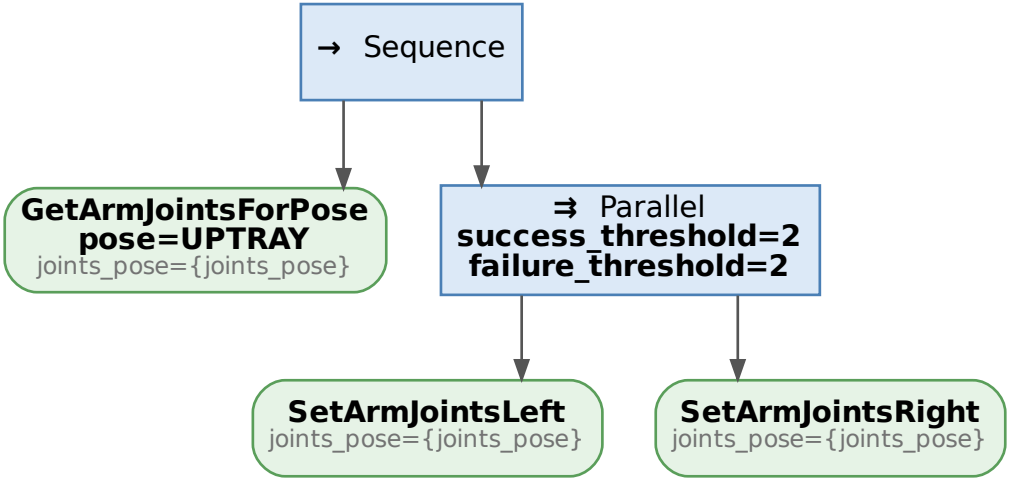
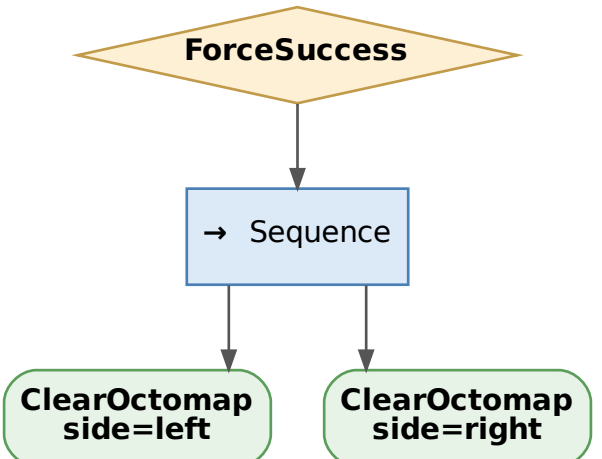
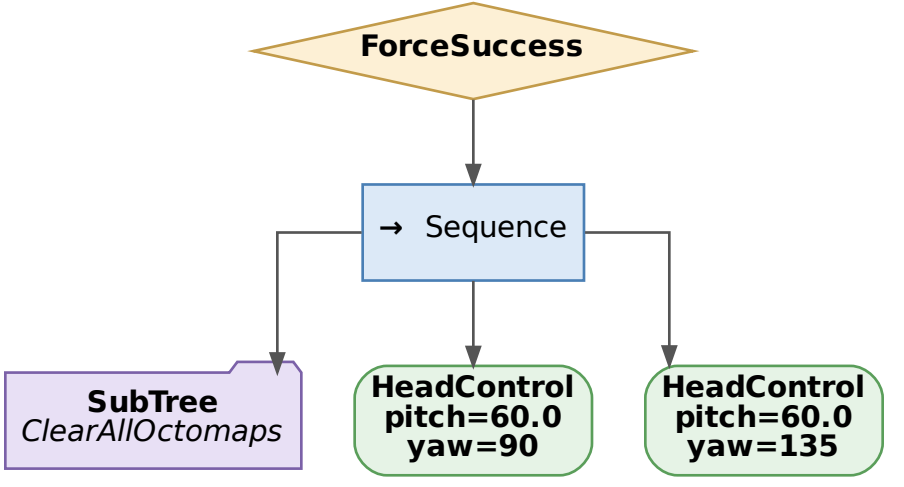

→ Sequence
SubTree
FullCompetitionMode

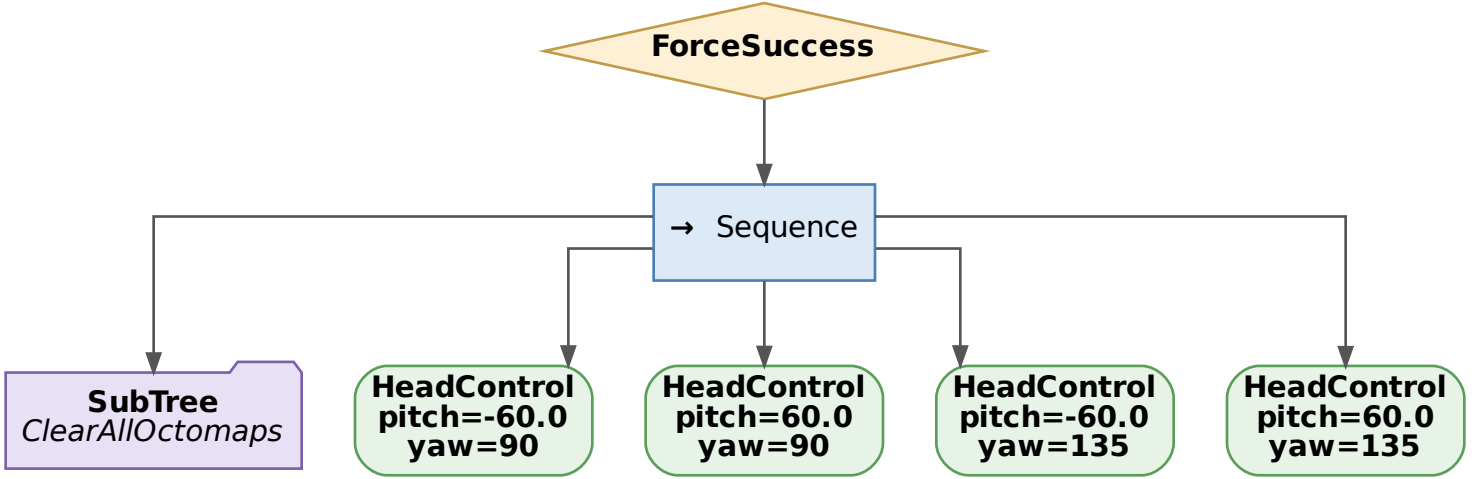

Repeat
num_cycles=-1
→ Sequence
SubTree
GetInRestPose
SubTree
DoYourTrick

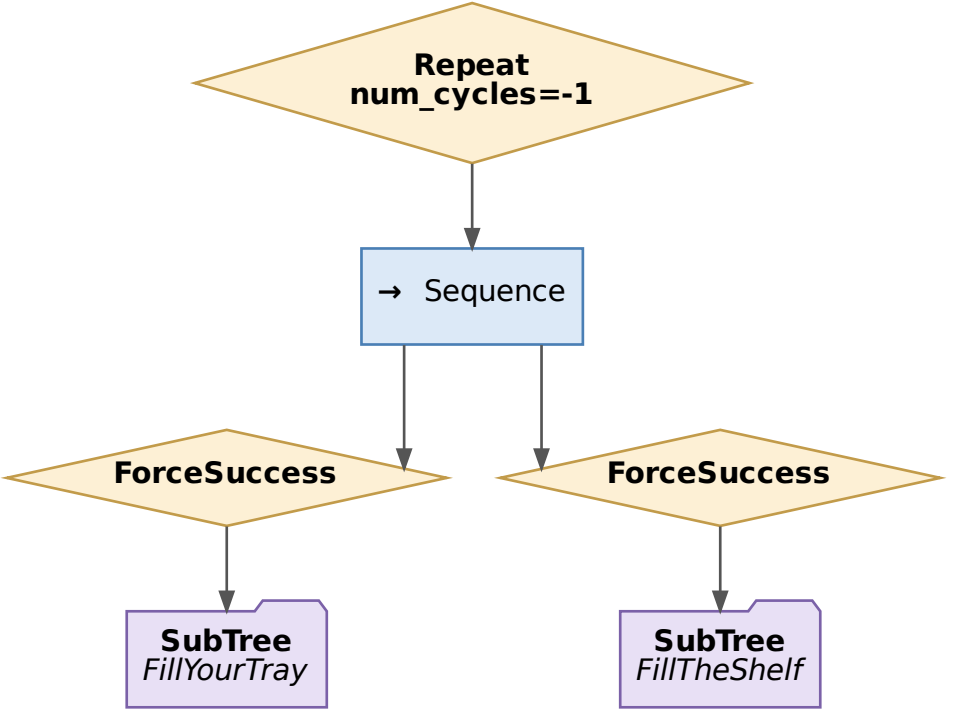

→ Sequence
GetArmJointsForPose
pose=REST
joints_pose={joints_pose}
⇉ Parallel
success_threshold=3
failure_threshold=3
→ Sequence
SetArmJointsLeft
joints_pose={joints_pose}
SetArmJointsRight
joints_pose={joints_pose}
HeadControl
pitch=0
yaw=90
TorsoControl
relative_height=0.05

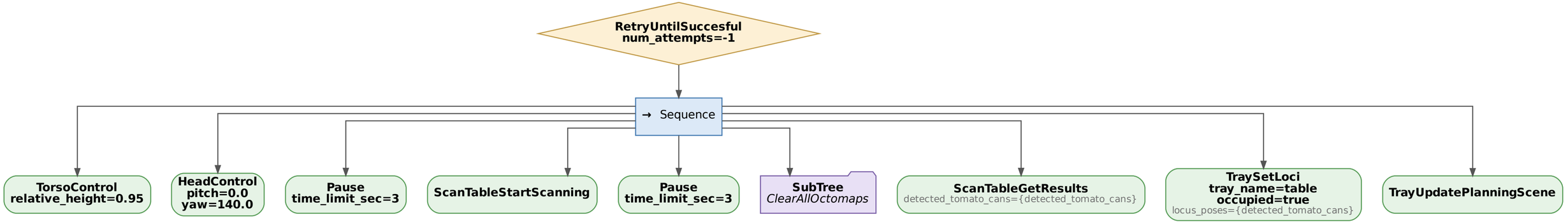

→ Sequence
GetArmJointsForPose
pose=UPTRAY
joints_pose={joints_pose}
⇉ Parallel
success_threshold=2
failure_threshold=2
SetArmJointsLeft
joints_pose={joints_pose}
SetArmJointsRight
joints_pose={joints_pose}

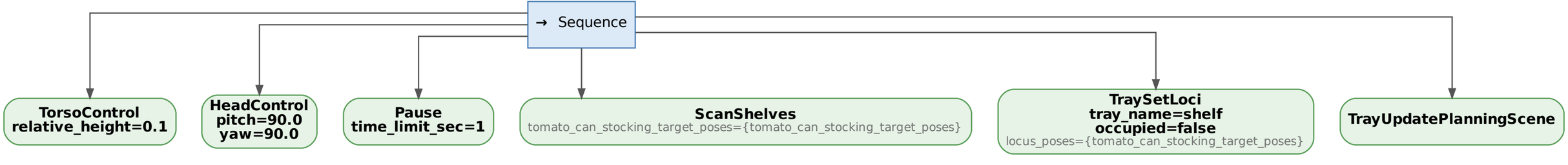

ForceSuccess
→ Sequence
ClearOctomap
side=left
ClearOctomap
side=right

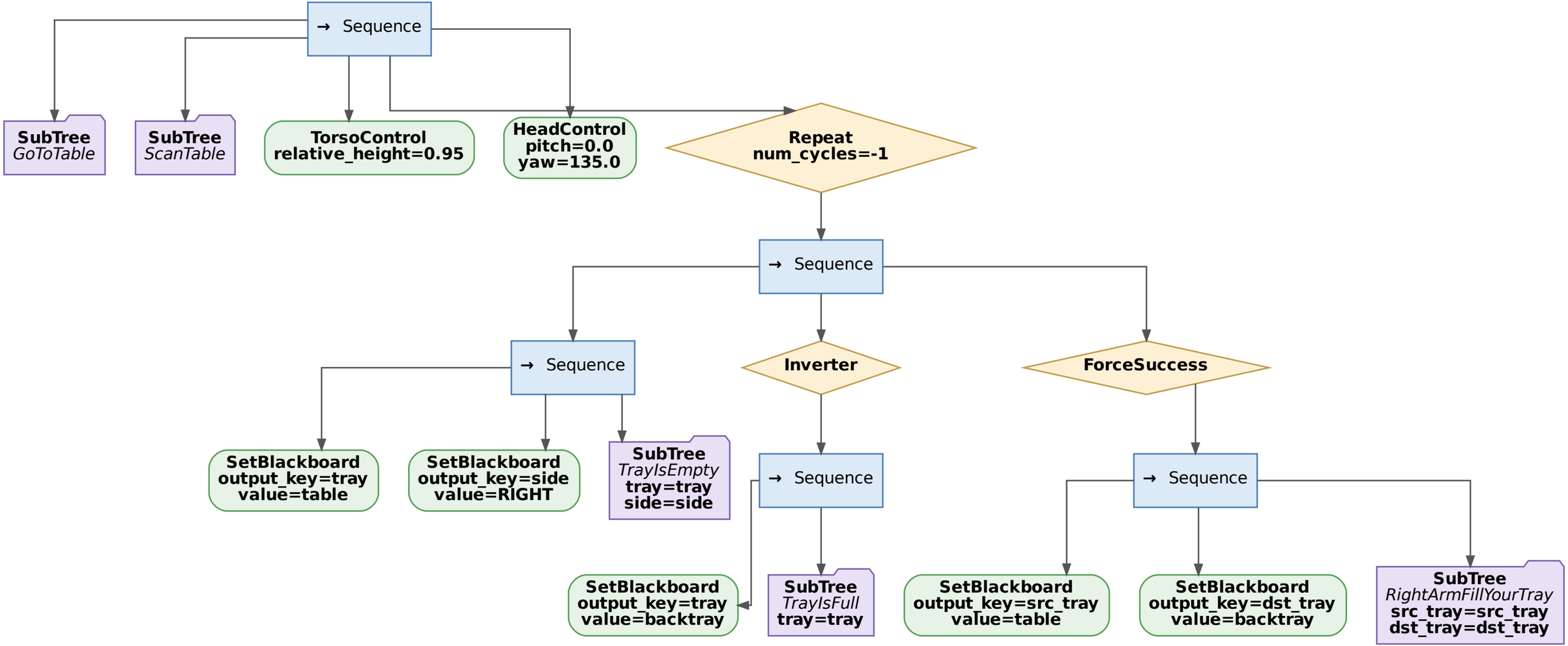

ForceSuccess
→ Sequence
SubTree
ClearAllOctomaps
HeadControl
pitch=60.0
yaw=90
HeadControl
pitch=60.0
yaw=135

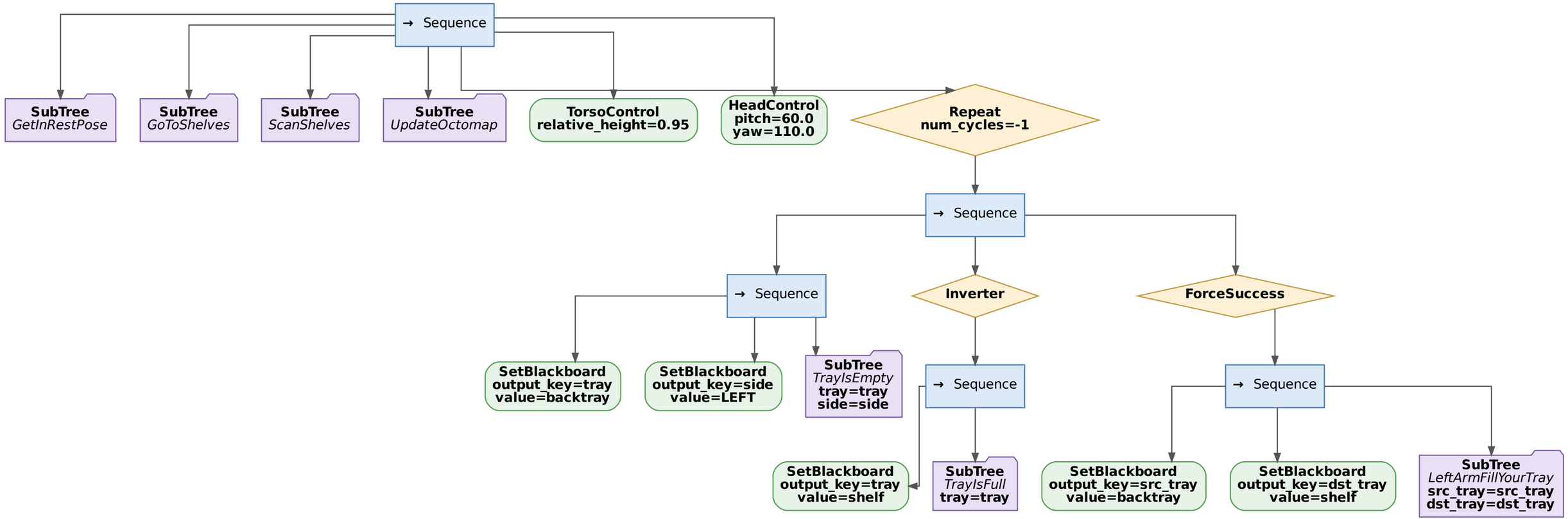

ForceSuccess
→ Sequence
SubTree
ClearAllOctomaps
HeadControl
pitch=-60.0
yaw=90
HeadControl
pitch=60.0
yaw=90
HeadControl
pitch=-60.0
yaw=135
HeadControl
pitch=60.0
yaw=135

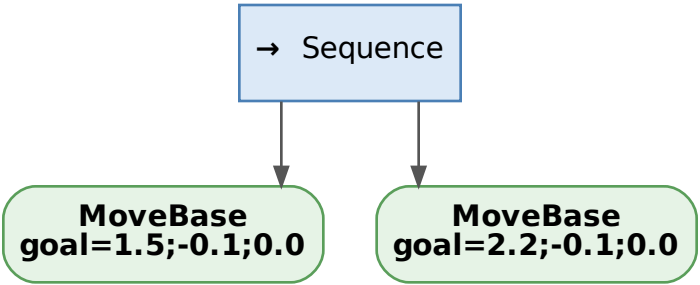

Repeat
num_cycles=-1
→ Sequence
ForceSuccess
ForceSuccess
SubTree
FillYourTray
SubTree
FillTheShelf

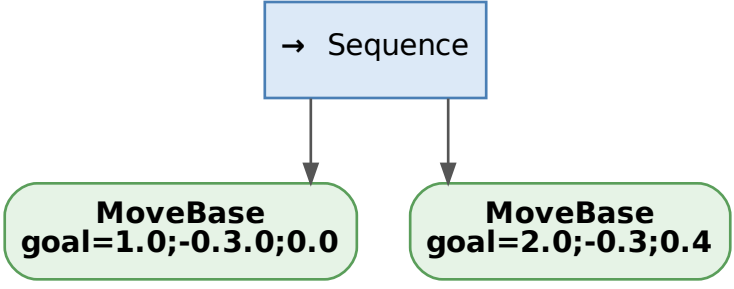

RetryUntilSuccesful
num_attempts=-1
→ Sequence
TorsoControl
relative_height=0.95
HeadControl
pitch=0.0
yaw=140.0
Pause
time_limit_sec=3
ScanTableStartScanning
Pause
time_limit_sec=3
SubTree
ClearAllOctomaps
ScanTableGetResults
detected_tomato_cans={detected_tomato_cans}
TraySetLoci
tray_name=table
occupied=true
locus_poses={detected_tomato_cans}
TrayUpdatePlanningScene

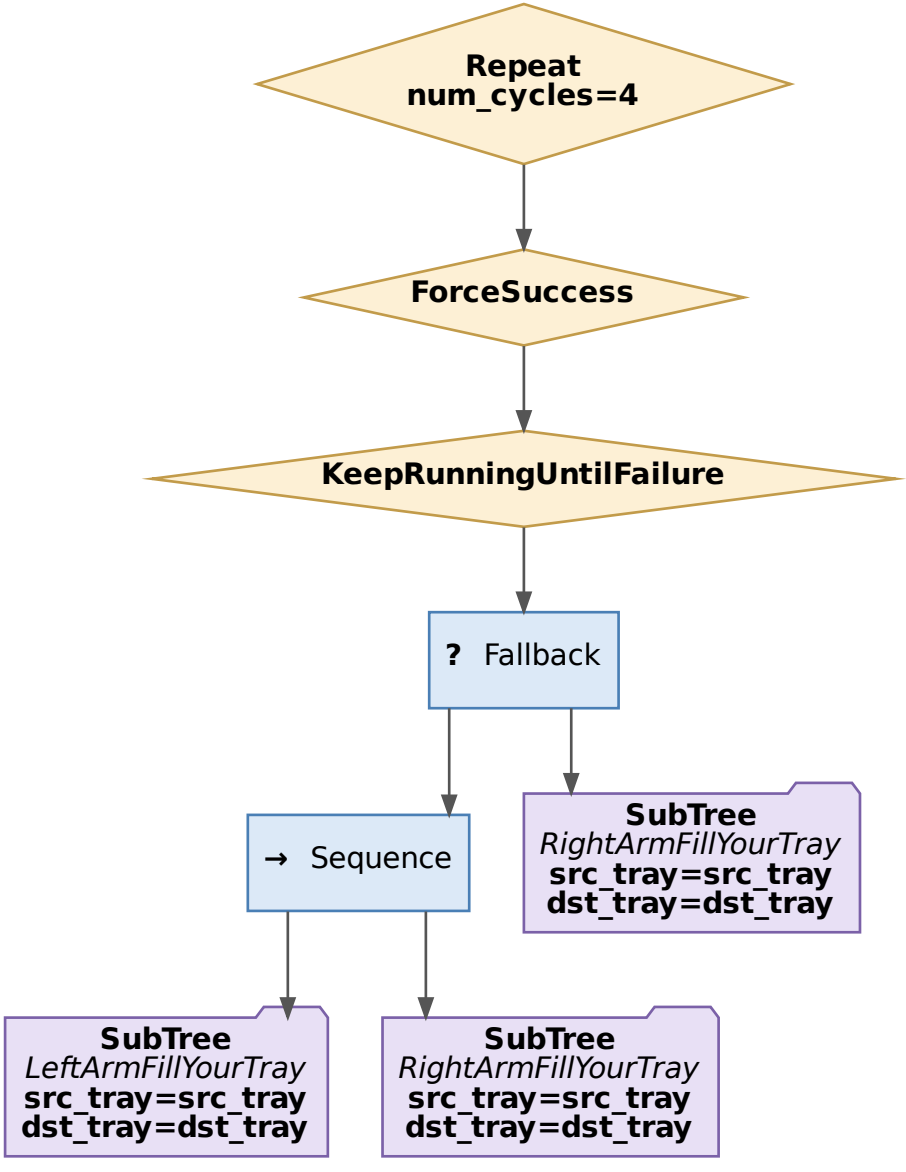

→ Sequence
TorsoControl
relative_height=0.1
HeadControl
pitch=90.0
yaw=90.0
Pause
time_limit_sec=1
ScanShelves
tomato_can_stocking_target_poses={tomato_can_stocking_target_poses}
TraySetLoci
tray_name=shelf
occupied=false
locus_poses={tomato_can_stocking_target_poses}
TrayUpdatePlanningScene

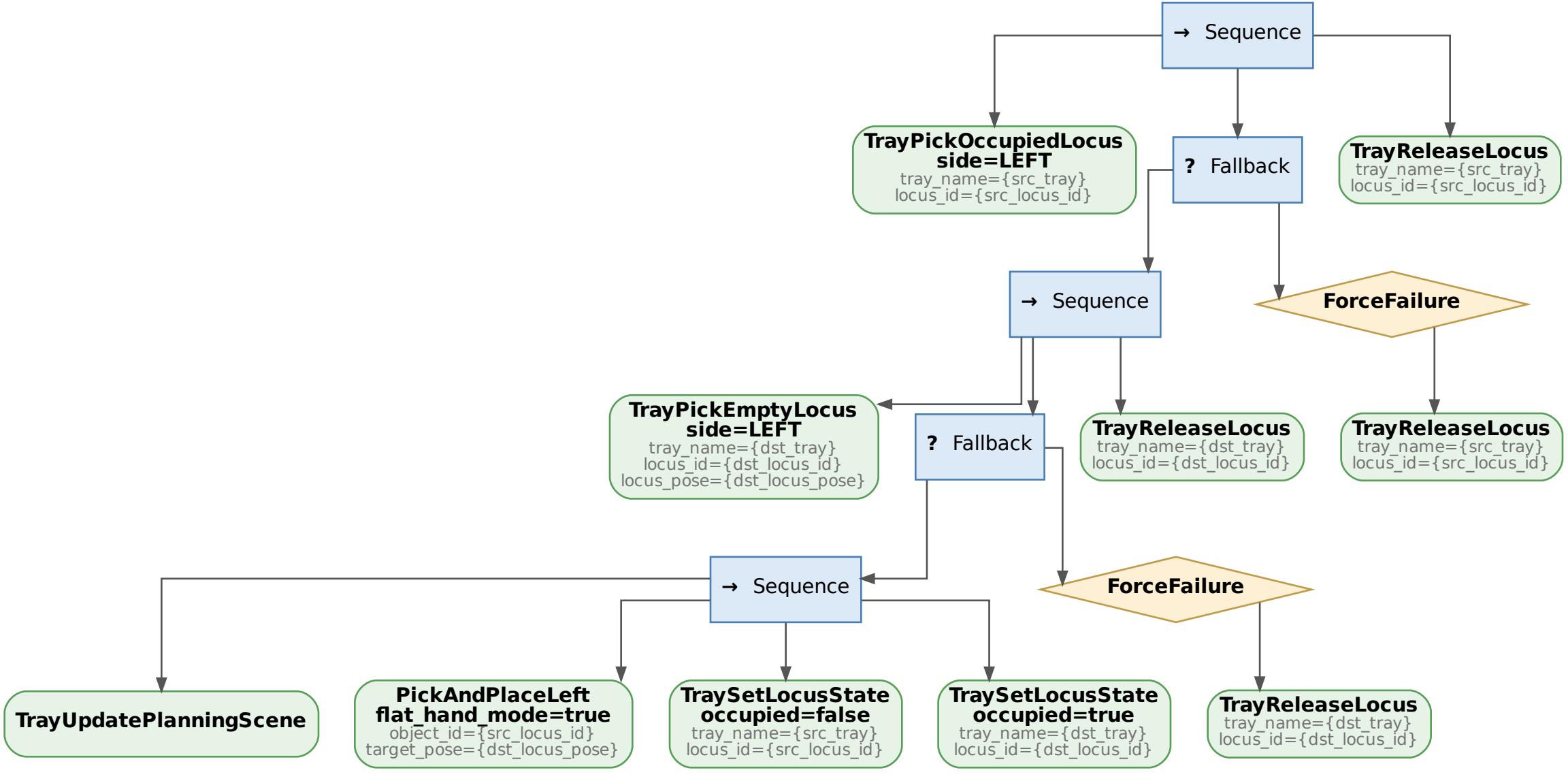

→ Sequence
SubTree
GoToTable
SubTree
ScanTable
TorsoControl
relative_height=0.95
HeadControl
pitch=0.0
yaw=135.0
Repeat
num_cycles=-1
→ Sequence
→ Sequence
Inverter
ForceSuccess
SetBlackboard
output_key=tray
value=table
SetBlackboard
output_key=side
value=RIGHT
SubTree
TrayIsEmpty
tray=tray
side=side
→ Sequence
→ Sequence
SetBlackboard
output_key=tray
value=backtray
SubTree
TrayIsFull
tray=tray
SetBlackboard
output_key=src_tray
value=table
SetBlackboard
output_key=dst_tray
value=backtray
SubTree
RightArmFillYourTray
src_tray=src_tray
dst_tray=dst_tray

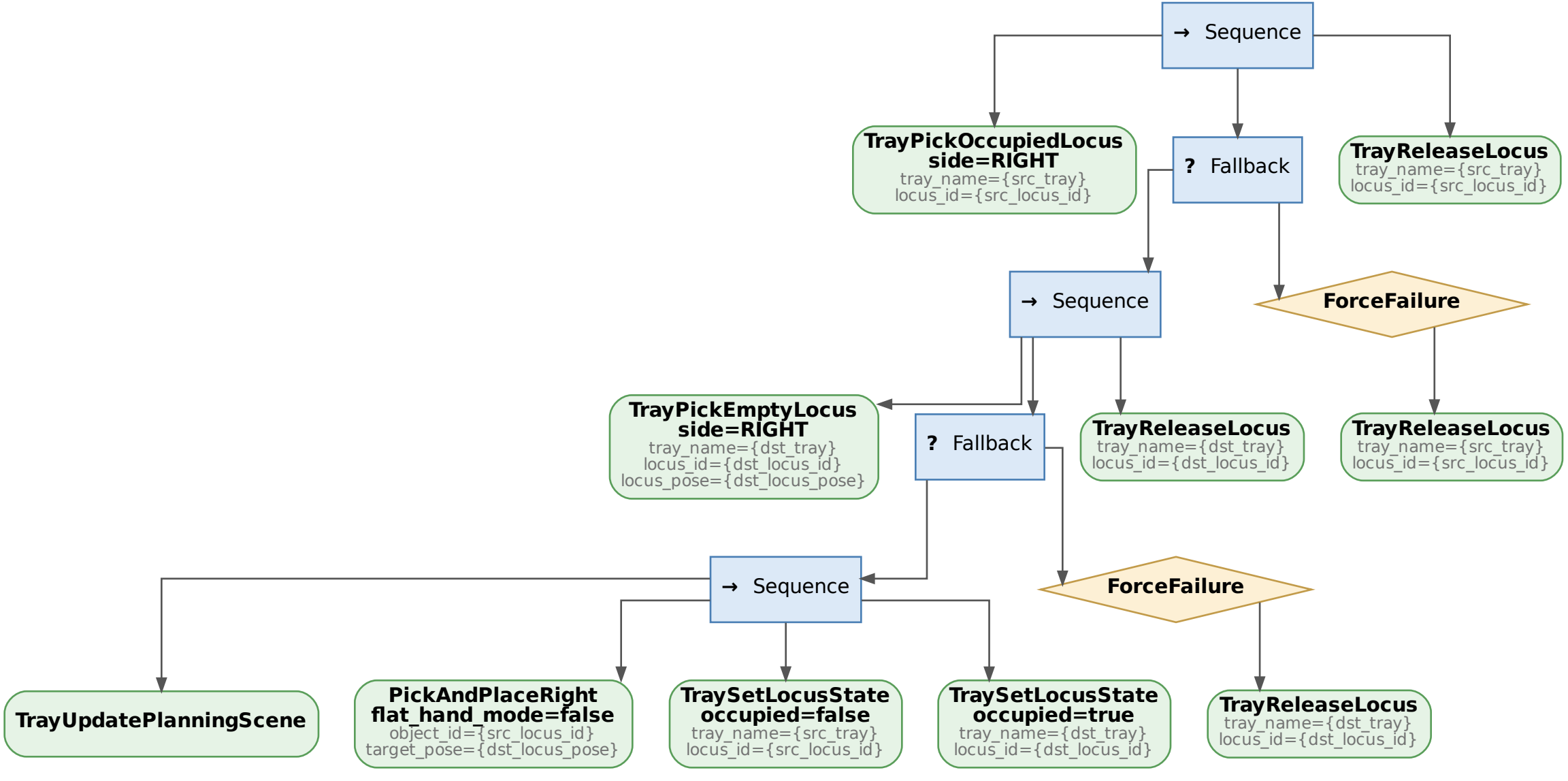

→ Sequence
SubTree GetInRestPose
SubTree GoToShelves
SubTree ScanShelves
SubTree UpdateOctomap
TorsoControl relative_height=0.95
HeadControl pitch=60.0 yaw=110.0
Repeat num_cycles=-1
→ Sequence
→ Sequence
Inverter
ForceSuccess
SetBlackboard output_key=tray value=backtray
SetBlackboard output_key=side value=LEFT
SubTree TrayIsEmpty tray=tray side=side
→ Sequence
→ Sequence
SetBlackboard output_key=tray value=shelf
SubTree TrayIsFull tray=tray
SetBlackboard output_key=src_tray value=backtray
SetBlackboard output_key=dst_tray value=shelf
SubTree LeftArmFillYourTray src_tray=src_tray dst_tray=dst_tray

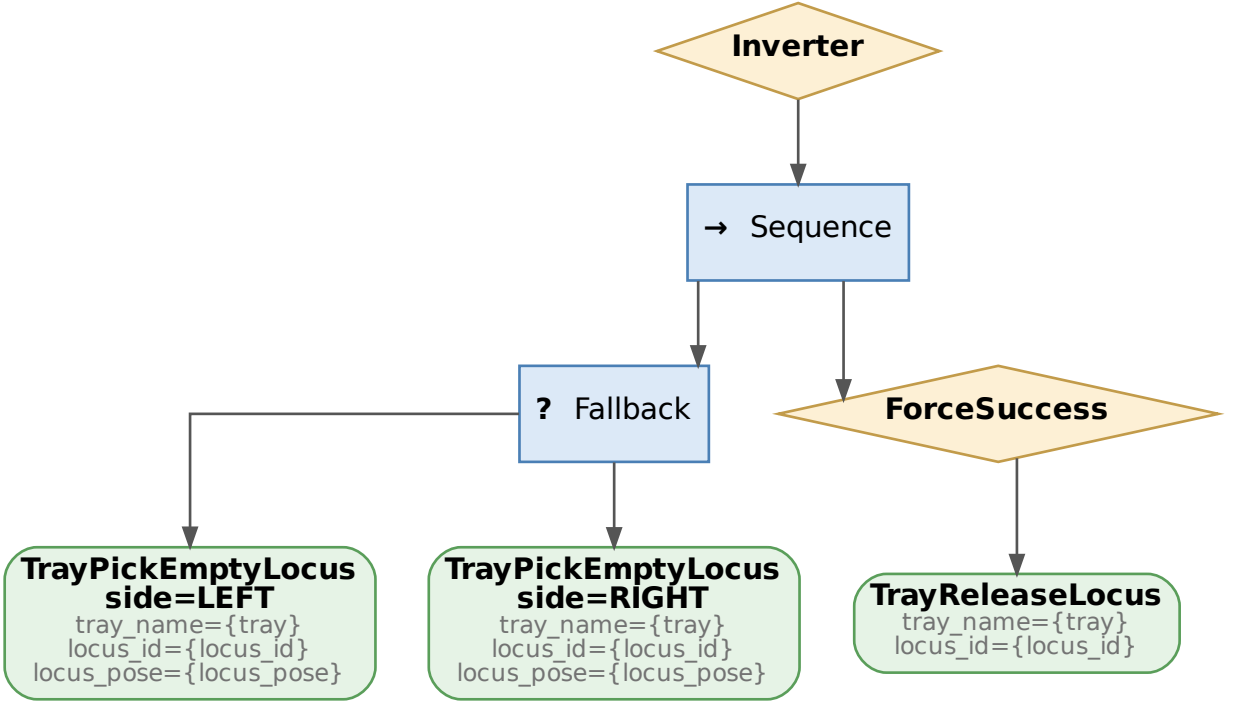

→ Sequence
MoveBase
goal=1.5;-0.1;0.0
MoveBase
goal=2.2;-0.1;0.0

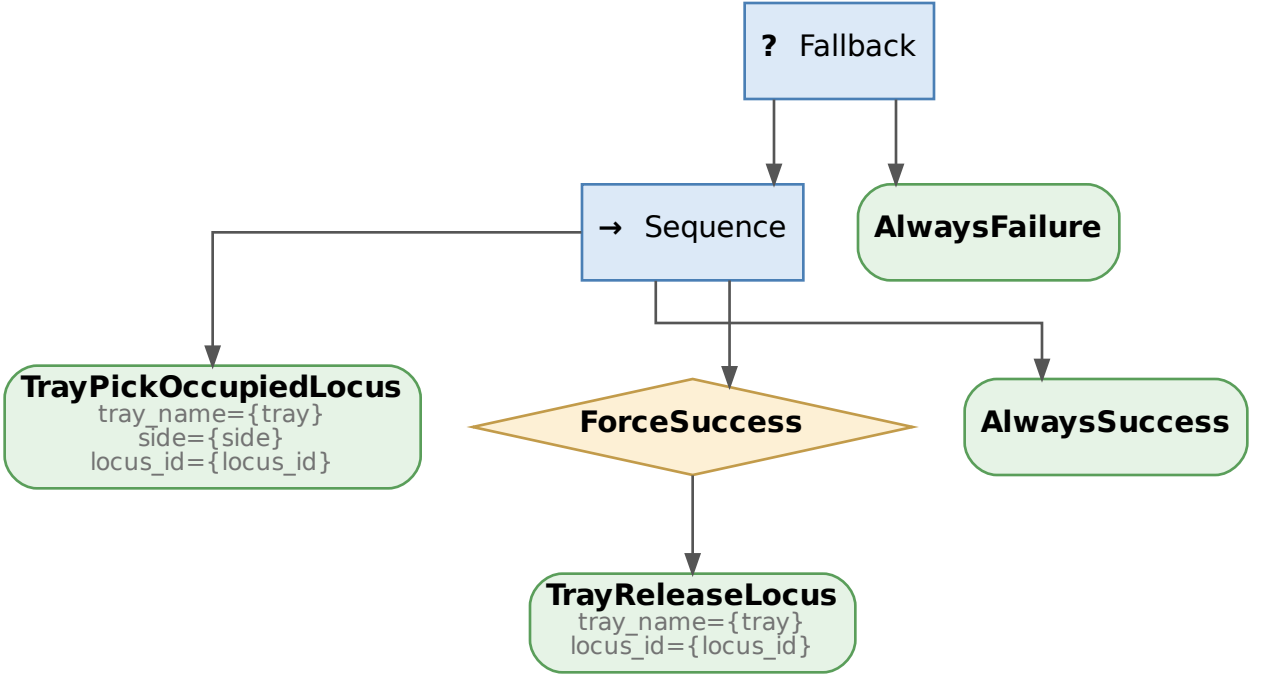

→ Sequence
MoveBase
goal=1.0;-0.3.0;0.0
MoveBase
goal=2.0;-0.3;0.4


Repeat
num_cycles=4
ForceSuccess
KeepRunningUntilFailure
? Fallback
→ Sequence
SubTree
RightArmFillYourTray
src_tray=src_tray
dst_tray=dst_tray
SubTree
LeftArmFillYourTray
src_tray=src_tray
dst_tray=dst_tray
SubTree
RightArmFillYourTray
src_tray=src_tray
dst_tray=dst_tray


→ Sequence
TrayPickOccupiedLocus
side=LEFT
tray_name={src_tray}
locus_id={src_locus_id}
? Fallback
TrayReleaseLocus
tray_name={src_tray}
locus_id={src_locus_id}
→ Sequence
ForceFailure
TrayPickEmptyLocus
side=LEFT
tray_name={dst_tray}
locus_id={dst_locus_id}
locus_pose={dst_locus_pose}
? Fallback
TrayReleaseLocus
tray_name={dst_tray}
locus_id={dst_locus_id}
TrayReleaseLocus
tray_name={src_tray}
locus_id={src_locus_id}
→ Sequence
ForceFailure
TrayUpdatePlanningScene
PickAndPlaceLeft
flat_hand_mode=true
object_id={src_locus_id}
target_pose={dst_locus_pose}
TraySetLocusState
occupied=false
tray_name={src_tray}
locus_id={src_locus_id}
TraySetLocusState
occupied=true
tray_name={dst_tray}
locus_id={dst_locus_id}
TrayReleaseLocus
tray_name={dst_tray}
locus_id={dst_locus_id}


→ Sequence
TrayPickOccupiedLocus
side=RIGHT
tray_name={src_tray}
locus_id={src_locus_id}
? Fallback
TrayReleaseLocus
tray_name={src_tray}
locus_id={src_locus_id}
→ Sequence
ForceFailure
TrayPickEmptyLocus
side=RIGHT
tray_name={dst_tray}
locus_id={dst_locus_id}
locus_pose={dst_locus_pose}
? Fallback
TrayReleaseLocus
tray_name={dst_tray}
locus_id={dst_locus_id}
TrayReleaseLocus
tray_name={src_tray}
locus_id={src_locus_id}
→ Sequence
ForceFailure
TrayUpdatePlanningScene
PickAndPlaceRight
flat_hand_mode=false
object_id={src_locus_id}
target_pose={dst_locus_pose}
TraySetLocusState
occupied=false
tray_name={src_tray}
locus_id={src_locus_id}
TraySetLocusState
occupied=true
tray_name={dst_tray}
locus_id={dst_locus_id}
TrayReleaseLocus
tray_name={dst_tray}
locus_id={dst_locus_id}


Inverter
→ Sequence
? Fallback
ForceSuccess
TrayPickEmptyLocus
side=LEFT
tray_name={tray}
locus_id={locus_id}
locus_pose={locus_pose}
TrayPickEmptyLocus
side=RIGHT
tray_name={tray}
locus_id={locus_id}
locus_pose={locus_pose}
TrayReleaseLocus
tray_name={tray}
locus_id={locus_id}


? Fallback
→ Sequence
AlwaysFailure
TrayPickOccupiedLocus
tray_name={tray}
side={side}
locus_id={locus_id}
ForceSuccess
AlwaysSuccess
TrayReleaseLocus
tray_name={tray}
locus_id={locus_id}